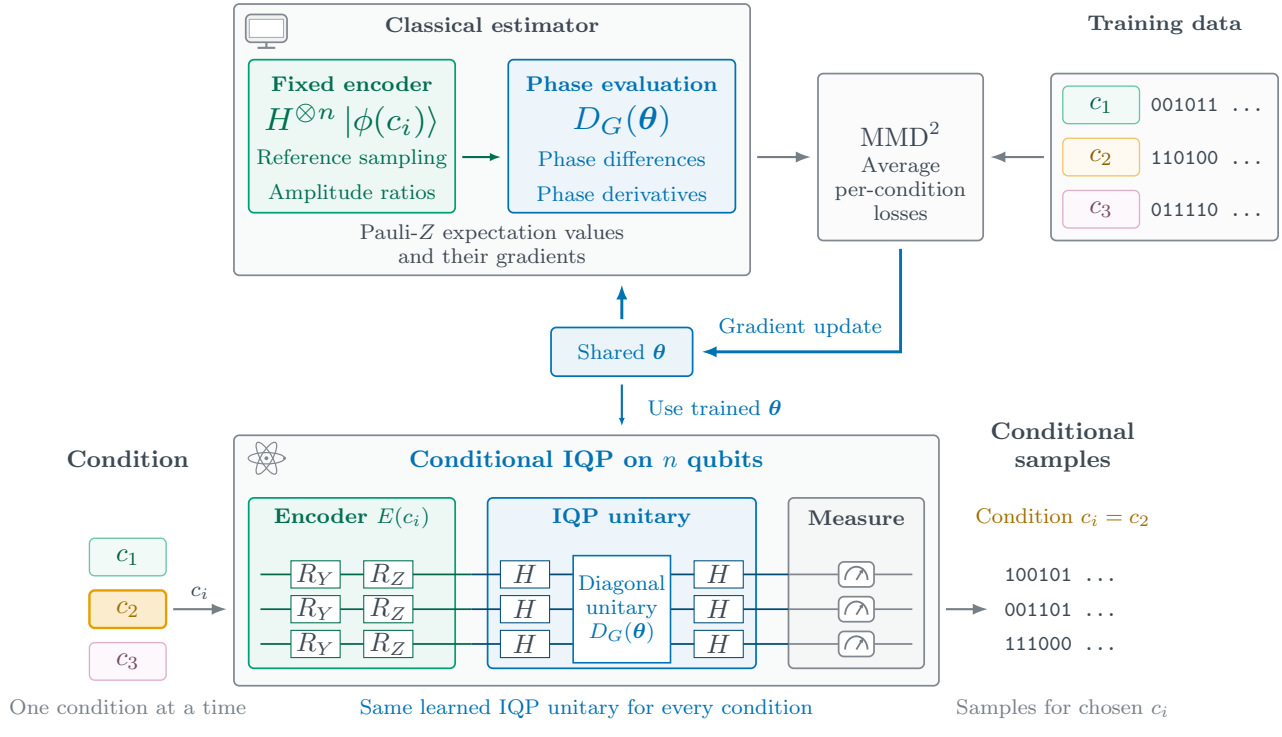

Classical estimator
Fixed encoder
$H^{\otimes n}\,|\phi(c_i)\rangle$
Reference sampling
Amplitude ratios
Phase evaluation
$D_G(\boldsymbol{\theta})$
Phase differences
Phase derivatives
Pauli-$Z$ expectation values
and their gradients
Training data
$c_1$ 001011 ...
$c_2$ 110100 ...
$c_3$ 011110 ...
$\mathrm{MMD}^2$
Average
per-condition
losses
Shared $\boldsymbol{\theta}$
Gradient update
Use trained $\boldsymbol{\theta}$
Condition
$c_1$
$c_2$
$c_3$
$c_i$
Conditional IQP on $n$ qubits
Encoder $E(c_i)$
$R_Y$ $R_Z$
IQP unitary
$H$
Diagonal
unitary
$D_G(\boldsymbol{\theta})$
Measure
Conditional
samples
Condition $c_i = c_2$
100101 ...
001101 ...
111000 ...
One condition at a time
Same learned IQP unitary for every condition
Samples for chosen $c_i$